

Arbitrum Grant Application - DecaFlow (Natural Rewrite)

Date: January 14, 2026

Program: New Protocols & Ideas

Amount: \$50,000 USD

Website: <https://decaflow.io>

GitHub: <https://github.com/affidexlab/new>

Q1: Project Details (Brief & Concise)

DecaFlow is privacy infrastructure for Arbitrum. We enable any protocol to offer MEV-protected transactions through a simple SDK integration.

The core idea: Instead of every protocol building MEV protection from scratch (which takes 4-8 weeks and deep expertise), they integrate our SDK in under an hour. We handle all the complexity - CoW Protocol routing, AI-powered risk analysis, optimal execution.

Think of it like Chainlink for privacy. Chainlink provides price feeds as infrastructure. We provide privacy routing as infrastructure.

Three main components: 1. **Privacy SDK** - Solidity, TypeScript, and Python libraries that protocols integrate 2. **AI Routing Engine** - Analyzes each trade and decides if MEV risk is high enough to justify privacy route 3. **Public MEV Dashboard** - Free analytics showing MEV extraction across Arbitrum

Current status: Live on Arbitrum mainnet with basic privacy routing working. Grant funds: SDK development, AI model training, protocol integrations (GMX, Camelot, etc).

Q2: Innovation & Value for Arbitrum

The Problem

Right now, Arbitrum doesn't have unified privacy infrastructure. Every protocol that wants MEV protection has to build it themselves. This is wasteful - it's like if every DeFi protocol had to build their own price oracle instead of using Chainlink.

The current options are: - Build MEV protection yourself (expensive, time-consuming, hard) - Hope users don't get sandwiched (they do, and it's bad UX) - Ignore the problem (limits institutional adoption)

Also, Arbitrum's Timeboost tech is underutilized. It's cutting-edge stuff but there aren't many production integrations yet.

Our Solution

1. First Composable Privacy Primitive on Arbitrum

We're not building another DEX. We're building infrastructure that makes ALL Arbitrum DEXs and protocols better.

Any protocol can integrate privacy in under an hour:

```
import "@decaflow/privacy-sdk/contracts/PrivacyRouter.sol";

contract MyDeFiProtocol {
    function swapWithPrivacy(address tokenIn, address tokenOut, uint256 amount) external {
        privacyRouter.executePrivateSwap(tokenIn, tokenOut, amount, msg.sender);
    }
}
```

That's it. Now that protocol offers MEV-protected swaps to their users.

2. AI-Powered Intelligent Routing

Current privacy solutions are dumb. They're either always-on (slow even when unnecessary) or always-off (get wrecked by MEV bots).

We built an AI that looks at each trade and says: "Is this risky enough to justify the 12-second delay of privacy routing?"

For example: - Small \$500 swap at 3 AM? Low risk. Route directly (fast). - Big \$100K swap at 2 PM during high gas? Critical risk. Route through CoW Protocol (MEV-protected).

The AI learns from Arbitrum transaction history to get better over time.

3. Timeboost Integration & Research

Arbitrum's Timeboost is unique - no other L2 has this. We want to be the first protocol to combine Timeboost with privacy routing.

Research questions: - Can Timeboost accelerate CoW Protocol settlements? - What's the cost/benefit of bidding for "fast privacy"? - When should the AI recommend this strategy?

We'll publish a research paper with our findings and share a prototype implementation.

4. Public Good: MEV Analytics Dashboard

Free dashboard showing: - How much MEV is being extracted on Arbitrum - Which token pairs are high risk - Personal wallet MEV exposure tracking

This raises awareness and incentivizes protocols to adopt privacy solutions.

Why This is New

No one else is doing this on Arbitrum: - CoW Swap exists but it's a consumer app, not infrastructure/SDK - 1inch aggregates prices but doesn't protect against MEV - OpenMEV exists but requires complex RPC-level changes

We're the first "privacy-as-infrastructure" on Arbitrum.

Value for Arbitrum

Competitive advantage: Makes Arbitrum the most MEV-resistant L2. Base, Optimism, zkSync don't have this.

Attracts users: Large traders currently avoid L2s because of MEV exposure. This solves that.

Ecosystem growth: Each protocol that integrates brings their users to Arbitrum-native privacy.

Showcases Timeboost: Demonstrates real-world use case for Arbitrum's cutting-

edge tech.

Measurable impact: Every dollar of MEV saved is direct value to Arbitrum users.
Target: \$1M+ saved by month 6.

Q3: Target Audience

Primary: DeFi Developers

Developers building on Arbitrum who want to offer MEV protection without spending weeks building it themselves. There are 500+ active protocol teams on Arbitrum. Our SDK makes their life way easier.

Primary: Large Traders

Wallets with \$50K+ positions who currently lose 5-10% to MEV on big trades. These are the folks who often use CEXs because DEXs are too risky. There are roughly 50,000 whale wallets on Arbitrum.

Secondary: Regular DeFi Users

Once protocols integrate our SDK, regular users automatically benefit from MEV protection without doing anything. 100,000+ monthly active DEX traders on Arbitrum.

Secondary: Researchers

Blockchain researchers and data scientists who want MEV data. We're providing the first comprehensive public MEV analytics for Arbitrum.

Q4: Team Experience & Completeness

Team is Complete - No Hiring Needed

Utibe Samuel - Founder & Project Lead

I've been in Web3 for 6+ years: - **CreedTech Group (2019-2021)**: Co-founded a blockchain dev agency. Managed a distributed team of smart contract developers. We built DeFi protocols, NFT projects, and DAO governance systems for various clients. I oversaw the Solidity audits and deployments. - **Blockroll (2021-2022)**: Ran partnerships for a Web3 ecosystem. Led campaigns that grew the ecosystem by 25%. - **Affidex Lab (2022-2024)**: Business development for a blockchain + AI incubator. - **DecaFlow (2024-present)**: Founded this after getting sandwich attacked one too many times trading on L2s.

My role: Product strategy, business development, partnerships, grant execution, community management.

Edidiong Samuel - CTO & Lead Developer

My brother, 5+ years of blockchain and full-stack development: - **TransactX (2020-2022)**: Led the technical side of a fintech API. Built a system handling 50,000+ users and 15,000+ daily transactions with 99.9% uptime. Not crypto but proves I can build production-grade systems. - **VestPad (2021-2022)**: Smart contract developer for a DeFi launchpad. Wrote Solidity contracts deployed on Ethereum and BSC. Used The Graph for indexing. - **Swap2Naira (2022)**: Backend developer for a DEX. Laravel + React stack. - **DecaFlow (2024-present)**: Built everything - smart contracts, CoW Protocol integration, cross-chain bridges, frontend, backend. All production code is

mine.

GitHub: <https://github.com/dakeem05> (76 public repos)

His role: All technical - smart contracts, SDK development, AI model implementation, security.

What We've Already Built (Proof We Can Deliver)

DecaFlow is live on Arbitrum mainnet right now: - Smart contracts deployed and verified - CoW Protocol integration functional - Cross-chain bridges working (CCIP, CCTP, Socket) - Frontend and backend infrastructure operational - GitHub: <https://github.com/affidexlab/new> (500+ commits, fully public)

We're not asking for money to build something theoretical. We've already built a complex multi-chain DEX. This grant scales it into ecosystem infrastructure.

References

We can provide references from: - Blockroll team (confirm my partnership work) - Early DecaFlow testers (confirm the protocol works) - Anyone can verify our GitHub activity and deployed contracts

Q5: Comparable Projects

CoW Swap (Ethereum/Gnosis Chain)

Best-in-class MEV protection using batch auctions. We actually use CoW Protocol as our privacy routing layer.

Difference: They're a consumer DEX app. We're infrastructure that ANY protocol can use. Also, we're Arbitrum-native with Timeboost integration.

1inch (Multi-chain)

DEX aggregator that finds best prices.

Difference: They optimize for price. We optimize for privacy. Different goals. Also, they don't have AI routing or Arbitrum-specific features.

OpenMEV (Manifold Finance)

MEV protection infrastructure.

Difference: Their integration requires RPC-level changes (complex). Ours is simple SDK integration. Plus we're Arbitrum-specific with Timeboost work.

Railgun (Multi-chain)

Privacy via zk-SNARKs.

Difference: Super complex UX (shielding/unshielding). Our privacy is transparent to users. Also focused on asset storage, not trading.

Within Arbitrum Ecosystem

GMX, Camelot, Radiant - these are potential integration partners, not competitors. Our value prop: We add privacy features to their protocols.

The Market Gap

There's NO unified privacy SDK on Arbitrum. Everything is either: - Consumer apps (not infrastructure) - Other chains (not Arbitrum-focused) - Complex to integrate (we're simple) - Static routing (we're AI-powered)

We fill that gap.

Q6: Current Stage

Already Deployed - Seeking Growth Funding

Status: Live on Arbitrum mainnet since December 20, 2025

What works right now: - Privacy routing via CoW Protocol - Smart contracts deployed on Arbitrum - Multi-chain support (Base, Arbitrum, Optimism, Polygon) - Cross-chain bridges (CCTP, CCIP, Socket) - Frontend with wallet integration - Backend API

Current usage (honest numbers): - ~100 early testers - ~500 test swaps executed - ~150 privacy swaps via CoW Protocol - ~\$150K volume

Why so small? We intentionally did a soft launch. We wanted to validate the tech works before marketing. No point driving users to an incomplete product.

What This Grant Funds (NOT Retroactive)

The grant funds NEW work only: 1. Privacy SDK (Solidity, TypeScript, Python libraries) - NOT YET BUILT 2. AI MEV prediction engine - data collection started, model NOT TRAINED 3. Protocol integrations (GMX, Camelot, +1) - talks initiated, NOT COMPLETE 4. MEV analytics dashboard - design done, NOT IMPLEMENTED 5. Timeboost research paper - NOT STARTED 6. Security audit - NOT STARTED

Post-grant plan: - Month 1-2: Launch SDK, train AI model - Month 3-4: Complete protocol integrations - Month 5-6: Launch public dashboard, finish audit

Target: Transform from "100 users" to "10,000+ users" by enabling ecosystem-wide adoption through integrations.

Q7: Previous Arbitrum Grants

No

This is our first Arbitrum grant application. We've self-funded development so far (~\$30K personal investment). No grants from Arbitrum DAO, Foundation, or ecosystem programs.

Q8: Other Blockchain Grants

No

No grants from Base, Optimism, Ethereum Foundation, or any other blockchain. 100% self-funded by the founding team.

Why apply to Arbitrum first? Largest L2 ecosystem, Timeboost technology is unique, best fit for privacy infrastructure.

Q9: Detailed Project Description

Executive Summary

We're building Arbitrum's first AI-powered privacy infrastructure. Any protocol can integrate MEV protection through our SDK. We handle the complexity, they get the benefits.

The Three Pillars

PILLAR 1: Privacy SDK

Problem: Every protocol builds MEV protection independently. Wasteful.

Solution: Open-source SDK that protocols integrate in under an hour.

For Solidity devs:

```
import "@decaflow/privacy-sdk/contracts/PrivacyRouter.sol";  
// Then call privacyRouter.executePrivateSwap(...)
```

For TypeScript devs:

```
import { DecaFlowPrivacy } from '@decaflow/privacy-sdk';  
const privacy = new DecaFlowPrivacy({network: 'arbitrum'});  
await privacy.executeSwap({...});
```

For Python devs:

```
from decaflow import PrivacyClient  
privacy = PrivacyClient(network='arbitrum')  
result = privacy.execute_swap(...)
```

Target integrations by month 6: - GMX (derivatives - privacy for collateral deposits) - Camelot (DEX - privacy toggle for large swaps) - Radiant or Vertex (lending/derivatives - privacy for deposits/withdrawals) - +2 community integrations

PILLAR 2: AI Routing

Problem: Privacy is always slow, direct swaps are always exposed. No intelligence.

Solution: AI analyzes each trade in <100ms and decides optimal route.

How it works: 1. User wants to swap 2. AI analyzes: swap size, liquidity depth, gas prices, time of day, mempool congestion, historical MEV patterns 3. AI outputs risk score (0-10) and route recommendation 4. High risk → privacy route. Low risk → direct route.

ML approach: XGBoost model trained on 50M+ Arbitrum swaps. Features: token volatility, swap size/liquidity ratio, gas correlation, arbitrage indicators, etc.

Target accuracy: >85% on validation set.

PILLAR 3: MEV Analytics Dashboard

Problem: Users don't realize they're losing money to MEV.

Solution: Free public dashboard at arbitrum-mev-dashboard.decaflow.io

Features: - Personal wallet MEV exposure tracking - Token pair risk scoring - Protocol-level analytics - Monthly ecosystem reports

This educates users and drives adoption of privacy solutions.

How Privacy Actually Works

1. User signs off-chain intent (not a transaction)
2. Intent goes to CoW Protocol's solver network (NOT public mempool)
3. Solvers compete to batch multiple intents together
4. Only final settlement hits blockchain
5. Result: No front-running because bots never saw the trade coming

The privacy comes from "dark orderflow" - transactions are hidden until after execution.

Problems We Solve

1. **Unified Infrastructure:** Stop duplicating work across protocols
2. **Intelligent Routing:** AI decides optimal path
3. **Timeboost Utilization:** First production integration
4. **MEV Visibility:** Dashboard educates ecosystem

Value for Arbitrum

- Attracts large traders avoiding L2s
- Makes all protocols better (network effect)
- Positions Arbitrum as most MEV-resistant L2
- Showcases Timeboost technology

Q10: Major Deliverables

Milestone 1 (Month 1-2): \$20,000

Deliverables: - Privacy SDK smart contracts deployed on Arbitrum (PrivacyRouter.sol, MEVOracle.sol) - TypeScript SDK published to NPM - Python SDK published to PyPI - AI MEV prediction engine trained (>85% accuracy, <100ms latency) - Prediction API live with documentation - 3 video tutorials for developers - Developer playground

KPIs: - AI accuracy: >85% - SDK downloads: 50+ - API uptime: >99%

Milestone 2 (Month 3-4): \$15,000

Deliverables: - GMX integration live on mainnet - Camelot integration live on mainnet - Third protocol integration (Radiant or Vertex) - 3 integration case studies published - Developer bounty program (\$5K pool)

KPIs: - Live integrations: 3+ - Privacy transactions: 500+ on-chain - Volume through privacy: \$5M+ - MEV saved: \$100K+

Milestone 3 (Month 4-5): \$10,000

Deliverables: - Public MEV dashboard launched - Real-time data indexing (1M+ Arbitrum swaps) - Public API for researchers - First monthly MEV report published

KPIs: - Dashboard monthly users: 1,000+ - Swaps indexed: 1M+ - Report readers: 5,000+

Milestone 4 (Month 6): \$5,000

Deliverables: - Timeboost research paper published (20+ pages) - Timeboost prototype on testnet - Security audit completed (Trail of Bits or Quantstamp) - Bug bounty program launched (\$50K pool) - Gas optimizations (20%+ reduction)

KPIs: - Paper views: 500+ - Audit: 0 unresolved critical findings - Bug bounty: Program live

Cumulative Success (Month 6): - \$1M+ MEV saved on Arbitrum - 5+ protocol integrations - 10,000+ unique users - \$25M+ volume - 500+ SDK downloads

All metrics are on-chain verifiable.

Q11: Arbitrum Ecosystem Alignment

DeFi Dominance

We make Arbitrum the most MEV-resistant L2. This is a competitive advantage over Base, Optimism, zkSync.

Attracts large traders who currently avoid L2s due to MEV exposure. Measurable: \$1M+ MEV saved by month 6.

Developer Tools

Privacy SDK reduces dev time from 4-8 weeks to under 1 hour. Free MEV risk API for all Arbitrum developers.

Measurable: 500+ SDK downloads, 20+ developers building with it.

Ecosystem Expansion

Attracts whales, institutions, privacy-conscious users. Each integrated protocol brings their users to Arbitrum.

Network effects: More integrations = more value for everyone.

Measurable: 10,000+ users by month 6, 500+ new-to-Arbitrum wallets.

Timeboost Program

First production integration of Timeboost with privacy routing. Research paper published for community.

Showcases Arbitrum's cutting-edge tech.

Measurable: 500+ research paper views, Timeboost value demonstrated.

How We Create Growth

Virtuous cycle: SDK → Protocol integrations → User adoption → More MEV data → Better AI → More value → More integrations.

Result: Arbitrum becomes "the privacy L2" with clear differentiation from competitors.

Q12: Grant Request

\$50,000 USD

We're requesting the full \$50K because the project spans three major technical pillars (SDK, AI Engine, MEV Dashboard). \$25K would require cutting essential components. Full \$50K enables complete ecosystem infrastructure.

Q13: Budget Breakdown

Category	Amount	Details
SDK Development	\$12,000 (24%)	Solidity contracts (\$4K), TypeScript SDK (\$3K), Python SDK (\$2.5K), docs (\$1.5K), video tutorials (\$1K)
AI MEV Engine	\$10,000 (20%)	Data collection (\$3K), ML training (\$3.5K), API development (\$2.5K), retraining pipeline (\$1K)
Protocol Integrations	\$8,000 (16%)	GMX integration (\$3K), Camelot (\$3K), third protocol (\$1.5K), developer bounties (\$500)
MEV Dashboard	\$7,000 (14%)	Frontend (\$3K), backend (\$2.5K), data indexing (\$1K), infrastructure 6mo (\$500)
Timeboost Research	\$5,000 (10%)	Research & analysis (\$2.5K), prototype (\$1.5K), community engagement (\$1K)
Security Audit	\$5,000 (10%)	Preliminary audit (\$4K), bug bounty setup (\$1K)
Documentation & DevRel	\$2,000 (4%)	Written docs (\$800), community support (\$600), outreach (\$600)
Infrastructure	\$1,000 (2%)	Hosting, RPC, database, domains for 6 months
TOTAL	\$50,000	100%

Notes: - No founder salaries (we work for equity) - Conservative rates (50-75% below market) - No retroactive funding - All costs are development/infrastructure only

Q14 & 15: Milestones with KPIs

See Milestone section (Q10) above for detailed breakdown

All milestones have: - Clear deliverables (binary verification) - Measurable KPIs (on-chain verifiable) - Specific deadlines (2-month chunks) - USD funding allocation

Q16: Project Timeline

6 months total from grant approval

- Month 1-2 (8 weeks): SDK & AI Foundation
- Month 3-4 (8 weeks): Protocol Integrations
- Month 4-5 (4 weeks): MEV Dashboard
- Month 6 (4 weeks): Security & Launch

Why this is realistic: - We already built a multi-chain DEX in 2 months - This timeline is conservative with buffer built in - Team commits full-time for 6 months - Parallel work where possible

Q17: Success Measurement (KPIs)

Primary Metrics (Most Important):

1. **MEV Saved:** \$1M+ by Month 6 (direct value to Arbitrum users, verifiable on-chain)
2. **Protocol Integrations:** 5+ live on mainnet (ecosystem adoption)
3. **User Adoption:** 10,000+ unique wallets (real users benefiting)
4. **Privacy Volume:** \$25M+ through privacy routes (scale of impact)

Technical Metrics:

5. **AI Accuracy:** >85% on validation set
6. **SDK Downloads:** 500+ (NPM + PyPI combined)
7. **API Usage:** 10,000+ calls/month
8. **Gas Efficiency:** <150K per privacy swap

Community Metrics:

9. **Dashboard MAU:** 5,000+ monthly active users
10. **Monthly Reports:** 5,000+ readers per report
11. **Media Coverage:** 3+ articles (CoinDesk, The Block, etc.)

Ecosystem Metrics:

12. **New Arbitrum Users:** 500+ new-to-Arbitrum wallets
13. **Developer Community:** 20+ actively building with SDK
14. **Integration Speed:** <2 weeks per protocol (proves SDK simplicity)

Security Metrics:

15. **Audit Results:** 0 unresolved critical findings
16. **Bug Bounty:** Program launched with \$50K pool

How We'll Report:

- Monthly transparency reports to Arbitrum DAO
- Public dashboard showing real-time metrics
- On-chain data (no trust needed, all verifiable)
- Screenshots, links, examples in each report

Success Threshold:

- Minimum (grant considered successful): \$500K MEV saved, 3 integrations, SDK functional
- Target: \$1M MEV saved, 5 integrations, 10K users
- Outstanding: \$2M MEV saved, 10 integrations, 20K users

Q18: Economic Sustainability

Revenue Model: Performance Fee (10% of MEV Saved)

Example: - Month 6: \$1M MEV saved → \$100K revenue - Monthly costs: ~\$16K (infrastructure, support, development) - Result: **Profitable by month 6**

Month 12 projection: - \$5M MEV saved → \$500K revenue - Monthly costs: ~\$20K - Result: **Highly profitable, no ongoing grants needed**

Fee Distribution: - 50% to DECA token stakers (protocol revenue share) - 30% to treasury (development, security, operations) - 20% to integrated protocols (revenue share incentive)

Break-Even: Month 9-10 (conservative estimate)

Post-Grant Revenue Streams:

1. Protocol fees (10% of MEV saved) - primary revenue
2. Enterprise SDK licenses (\$5K-\$25K/month per institution)
3. Premium API tiers (\$100-\$1K/month for high usage)
4. White-label solutions (\$50K-\$200K setup per chain)
5. Data licensing (\$500-\$5K/month for hedge funds)

Cost Optimization:

- Infrastructure scales efficiently (\$1K/month → \$500/month by month 12)
- Automation reduces support costs
- Open-source community contributions reduce dev costs

Long-Term Plan:

- Month 12-18: Token launch (\$DECA for governance and revenue share)
- Seed round: \$500K-\$2M additional funding
- Month 18+: DAO governance (community-owned public good)
- Treasury funded by ongoing fees

Result: Self-sustaining by month 6. No ongoing grants required.

Q19: Protocol Performance

Current Status: Live on Arbitrum mainnet since Dec 20, 2025

Deployments: - Smart contracts on Arbitrum (verified on Arbiscan) - Multi-chain: Base, Arbitrum, Optimism, Polygon - GitHub: <https://github.com/affidexlab/new> (500+ commits, public)

Technical Achievements: - CoW Protocol integration functional - Cross-chain bridges operational (CCTP, CCIP, Socket) - 50,000+ lines of code - 80%+ test coverage - Gas efficiency: 120K-180K gas per swap

Early Metrics (Pre-Marketing Phase): - ~100 early testers - ~500 test swaps executed - ~150 privacy swaps via CoW Protocol - 99%+ uptime - <5% failed transactions

Note on Limited Traction:

These numbers are small because we intentionally soft-launched. We wanted to validate the tech before marketing. Grant funds the SDK and integrations that enable scale.

Post-grant: 100 users → 10,000 users via protocol integrations.

Proven Capabilities:

Team built complex multi-chain protocol in 2 months. Demonstrates ability to deliver grant scope. Lower risk than unproven team with only ideas.

Future Tracking:

- Public MEV dashboard (Milestone 3)
- Monthly transparency reports

- On-chain metrics (fully verifiable)
 - Community can audit everything
-

Q20: Audit History & Security

Current Status: No external audit yet

Reason: Protocol launched Dec 2025. Waiting for code completion before audit (standard practice). Security audit budgeted in grant (Milestone 4, \$5K).

Security Measures Already Implemented:

- Solidity 0.8.20+ (latest stable version)
- OpenZeppelin libraries (industry standard)
- ReentrancyGuard, SafeERC20, Pausable patterns
- 80%+ test coverage (targeting 100% before audit)
- Multi-sig governance (Gnosis Safe)
- Extensive testnet validation

Planned Audit (Month 6):

- Firm: Trail of Bits or Quantstamp (reputable firms)
- Budget: \$5,000 (preliminary audit, covers core contracts)
- Scope: PrivacyRouter, MEVOracle, FeeManager
- Deliverable: Public audit report, all critical/high findings resolved before mainnet push

Bug Bounty Program (Month 6):

- Platform: Immunefi or Code4rena
- Pool: \$50,000 committed
- Rewards: Critical (\$10K-\$50K), High (\$5K-\$10K), Medium (\$1K-\$5K)
- Ongoing indefinitely

No bug bounty currently active - Launching after audit when code is production-ready and we're confident in security.

Q21: Composability

Yes - Highly Composable by Design

SDK Integration:

Any Arbitrum protocol can integrate privacy in under an hour. Just import our contract and call the function:

```
privacyRouter.executePrivateSwap(tokenIn, tokenOut, amount, msg.sender);
```

Examples: - GMX: Privacy for collateral deposits - Camelot: Privacy toggle for swaps - Radiant: Privacy for lending operations

AI Risk API:

Free public API for MEV risk scoring. Any protocol can query:

```
GET api.decaflow.io/v1/mev-risk?tokenIn=USDC&tokenOut=ETH&amount=1000000
```

Wallets can show MEV warnings, aggregators can route intelligently.

Data Composability:

Public MEV analytics that anyone can use: - Protocols can embed MEV stats in their UI - Arbitrum Foundation can showcase MEV resistance - Researchers get free data for academic work

Liquidity Composability:

We route through existing Arbitrum DEXs (Uniswap, Camelot, Balancer, SushiSwap). We aggregate liquidity, not compete for it.

Result: More volume for existing protocols, no fragmentation.

Non-Competitive:

We enhance protocols, not replace them. GMX, Camelot, Radiant all benefit from integration (more users + revenue share from fees).

Q22: Realistic Scope

Yes - Conservative and Well-Defined

Why We Can Deliver:

1. **Proven Team:** Already built multi-chain DEX (more complex than SDK). If we delivered that in 2 months, we can deliver SDK/AI/dashboard in 6 months.
2. **Conservative Timeline:** 6 months with buffer time. Month 6 is finalization and can absorb earlier delays.
3. **Sufficient Budget:** \$50K with detailed breakdown. We're 50-75% below market rates (founders work for equity, not salary).
4. **Clear Deliverables:** Every milestone has binary verification (SDK published? Yes/No. Audit done? Yes/No. No ambiguity.)
5. **Risk Mitigation:** Built-in buffers, fallback plans, must-have vs. nice-to-have separation.

Industry Benchmarks:

- CoW Swap: 8-10 months to launch
- 1inch: 12 months to launch
- Flashbots: 18 months to launch

Our 6-month timeline is faster because: smaller scope + existing working protocol + experienced team.

Not Overambitious:

- Only 3-5 integrations (not 100)
- Only 10,000 users (not 100,000)
- Only \$25M volume (not \$1B)
- Realistic KPIs based on proven benchmarks

Track Record:

We under-promise and over-deliver. DecaFlow protocol exceeded initial scope and timeline.

Flexibility:

Must-have (grant success if achieved): - 3+ integrations - SDK functional and

documented - Security audit complete

Nice-to-have (bonus if achieved): - 5+ integrations - 10,000+ users - \$1M+ MEV saved

We're confident in delivering the must-haves, optimistic about the nice-to-haves.

MILESTONES (Portal Format)

MILESTONE 1: SDK Development & AI Foundation

Details:

Privacy SDK smart contracts (PrivacyRouter.sol, MEVOracle.sol) deployed on Arbitrum mainnet and verified on Arbiscan. TypeScript SDK published to NPM at npmjs.com/package/@decaflow/privacy-sdk. Python SDK published to PyPI. AI MEV prediction engine trained on 50M+ Arbitrum transactions achieving >85% accuracy with <100ms inference latency. Prediction API live at api.decaflow.io with >99% uptime. Complete documentation: getting started guides for each language, API reference with 50+ code examples. 3 video tutorials published on YouTube. Developer playground at playground.decaflow.io for testing integrations.

KPIs: AI accuracy >85%, inference latency <100ms, gas cost <150K per swap, SDK downloads 50+ total, GitHub stars 25+, API uptime >99%, documentation 100% complete.

Verification: Contract addresses verifiable on Arbiscan. NPM package publicly available. PyPI package publicly available. Public GitHub repo with examples. API testable by anyone. YouTube playlist public.

Deadline: 8 weeks from grant approval

Funding: \$20,000 USD

MILESTONE 2: Protocol Integrations

Details:

GMX integration complete with privacy routing for collateral deposits, live on Arbitrum mainnet. Camelot integration complete with privacy toggle in swap UI, live on mainnet. Third protocol integration (Radiant or Vertex) for lending/derivatives privacy, live on mainnet. 3 detailed integration case studies published on Medium explaining technical implementation, benefits, and metrics. Developer bounty program launched with \$5K pool for community integrations. Weekly office hours for developers building integrations. Revenue share smart contracts deployed for fee distribution to integrated protocols.

KPIs: 3+ live mainnet integrations, 500+ privacy transactions verifiable on-chain, \$5M+ volume through privacy routes, \$100K+ MEV saved (calculated via price impact analysis), 5-10% of large trades using privacy mode, 500+ case study reads, 10+ bounty applicants, integration time <2 weeks per protocol.

Verification: Contract addresses on Arbiscan. Screenshots of privacy toggles in partner UIs. On-chain transaction examples (tx hashes). Published case studies with Medium links. Community feedback from partners.

Deadline: 8 weeks after Milestone 1 completion

Funding: \$15,000 USD

MILESTONE 3: MEV Analytics Dashboard & Public Goods

Details:

Public MEV dashboard launched at arbitrum-mev-dashboard.decaflow.io with features: wallet analytics (connect wallet to see personal MEV exposure), token pair risk scoring (e.g. "ETH/USDC: 6.2/10 risk"), protocol-level MEV analytics (which protocols suffer most MEV), ecosystem overview (total MEV extracted on Arbitrum). Backend infrastructure: real-time indexer processing new blocks, 1M+ Arbitrum swaps indexed, PostgreSQL + TimescaleDB for time-series data, public API endpoints, Redis caching for performance. Public API for researchers with free tier allowing CSV/JSON export. First comprehensive monthly MEV report published on Medium/Twitter with charts, analysis, trends.

KPIs: 1,000+ dashboard monthly active users (Google Analytics), 200+ unique wallets analyzed, 1M+ transactions indexed, <60 second data freshness, >99.5% API uptime, 5,000+ report readers, 50+ social media engagement, 20+ researchers using API.

Verification: Public dashboard URL accessible to anyone. Wallet connection test. Google Analytics (MAU). Published report with Medium link. API documentation public.

Deadline: 4 weeks after Milestone 2 completion

Funding: \$10,000 USD

MILESTONE 4: Timeboost Research, Security & Community Handover

Details:

Timeboost research paper (20+ pages) analyzing integration of Arbitrum Timeboost with CoW Protocol privacy routing. Methodology explained, 3 strategies evaluated (Timeboost+CoW, CoW only, Direct), cost-benefit analysis with real data, recommendations for optimal use. Paper published as PDF and posted to Arbitrum Forum for community feedback. Timeboost prototype: TimeboostBidder.sol smart contract deployed on Arbitrum testnet with benchmarking data. Security audit from Trail of Bits or Quantstamp covering PrivacyRouter, MEVOracle, FeeManager. Public audit report with 0 unresolved critical findings. Bug bounty program launched on Immunefi or Code4rena with \$50K pool. Gas optimization achieving 20%+ reduction (<120K gas per swap). Final comprehensive documentation: complete SDK reference, architecture diagrams, security best practices guide, FAQ. Community governance transition: DAO framework proposed, community forum launched, multisig setup, final transparency report covering all 6 months.

KPIs: 500+ research paper views, 50+ Arbitrum Forum comments on paper, 0 unresolved critical/high security findings, bug bounty program live and accepting submissions, 20%+ gas reduction vs baseline, 100% SDK documentation complete, DAO operational with first governance vote.

Cumulative Success Metrics (End of Month 6): \$1M+ MEV saved on Arbitrum (verifiable on-chain), 5+ protocol integrations live, 10,000+ unique users (wallet addresses), \$25M+ volume through privacy routes, 500+ SDK downloads total, 5,000+ dashboard MAU, 200+ GitHub stars, 3+ media articles about DecaFlow.

Verification: Research paper PDF link. Arbitrum Forum post link. Public audit report link. Bug bounty program link. Gas benchmarks published. Documentation site URL. Governance forum URL. Multisig address. Final transparency report.

Deadline: 4 weeks after Milestone 3 completion (End of Month 6)

Funding: \$5,000 USD

Final Notes

We appreciate the tough but fair feedback. We've addressed: - Technical depth on privacy mechanisms - Concrete traction data with context - Team crypto credentials and GitHub links - Clear Arbitrum alignment (not Base-focused) - Actionable, verifiable KPIs

We're a proven team (live protocol on mainnet) asking for growth capital to scale, not speculative funding for an idea.

Core ask: \$50K over 6 months to build privacy infrastructure (SDK, AI, integrations) that makes Arbitrum the most MEV-resistant L2.

Measurable outcome: \$1M+ MEV saved for Arbitrum users = 20x ROI on grant funds.

Thank you for considering our resubmission.

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GitHub: <https://github.com/affidexlab/new>